ONLINE EXAM CHEATING DETECTION USING CLASSIFICATION (SVM)

ACTIVITY REPORT PRESENTATION

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CHAPTER 1 : INTRODUCTION

- Online examinations face challenges in maintaining academic honesty.
- Students may use unfair methods such as tab switching or searching online.
- Machine Learning techniques help detect suspicious activities automatically.
- Support Vector Machine (SVM) classifies behavior as Normal or Suspicious.

CHAPTER 2 : PROBLEM STATEMENT

With the rise of online examinations, maintaining academic integrity has become challenging. The system logs data such as time spent per question ,frequency of the tab switching, mouse movements patterns, and answer submission timing .We should build a code to classify whether a student's behavior is "Normal" or "Suspicious, " helping to detect potential cheating.

CHAPTER 3 : OBJECTIVES

- Detect cheating behavior during online exams.
- Monitor activities such as tab switching and answer timing.
- Reduce manual invigilation effort.
- Improve fairness and integrity in examinations.
- Use machine learning for intelligent decision-making.

CHAPTER 4 : FEATURES COLLECTED BY THE SYSTEM

- Time spent per question.
- Tab switching frequency.
- Mouse movement patterns.
- Keyboard activity.
- Answer submission timing.
- Webcam monitoring and browser activity.

CHAPTER 5 : CLASSIFICATION AND SVM

•Classification predicts categories of data.

•Binary Classification: Normal or Suspicious behavior.

•SVM is a supervised machine learning algorithm.

SVM finds a hyperplane separating two classes.

CHAPTER 6 :WORKING PRINCIPLE & MATHEMATICAL MODEL

- Training data is analyzed to identify patterns.
- SVM creates a decision boundary.
- New student behavior is classified automatically.
- Decision Equation: $w \cdot x + b = 0$
- Result $> 0 \rightarrow$ Normal, Result $< 0 \rightarrow$ Suspicious

CHAPTER 7 : SYSTEM ARCHITECTURE

1. Data Collection
2. Data Preprocessing
3. Feature Extraction
4. Model Training
5. Classification
6. Alert Generation

CHAPTER 8 : ADVANTAGES & CHALLENGES

- Advantages: High accuracy, memory efficient, reliable performance.

- Handles complex cheating behavior patterns effectively.

- Challenges: Privacy issues, false detection, internet problems.

- Different hardware may affect data collection.

CHAPTER 9 : APPLICATIONS & TECHNOLOGIES USED

•Applications: School exams, entrance tests, certification exams.

•Technologies: Python, Scikit-learn, OpenCV, Pandas, NumPy.

•Flask/Django used for backend development.

CHAPTER 10 : CODE USED

```
# ONLINE EXAM CHEATING DETECTION USING SVM
# MULTIPLE STUDENTS INPUT AND OUTPUT
from sklearn import svm
# TRAINING DATASET
X = [
[45, 0, 120, 30] , [50, 1, 100, 25] , [48, 0, 110, 28] , [20, 8, 15, 5
] , [15, 10, 10, 3] , [18, 7, 20, 4] , [55, 1, 130, 35 ], [52, 2, 125, 30
] ,
 [17, 9, 18, 4] , [49, 0, 115, 27]
]
# LABELS
# 0 = NORMAL
# 1 = SUSPICIOUS
y = [0, 0, 0, 1, 1, 1, 0, 0, 1, 0]
# CREATE AND TRAIN MODEL
model = svm.SVC(kernel='linear')
model.fit(X, y)
# NUMBER OF STUDENTS
n = int(input("Enter number of students: "))
# INPUT AND OUTPUT
for i in range(n):
print("\nENTER DETAILS OF STUDENT", i + 1)
time_per_question = int(input("Average time per question:
"))
tab_switches = int(input("Number of tab switches: "))
mouse_movements = int(input("Mouse movements count:
"))
submission_time = int(input("Answer submission time: "))
```

CHAPTER 10 : CODE USED

```
# STUDENT DATA
student = [[
time_per_question,
tab_switches,
mouse_movements,
submission_time
]]
# PREDICTION
result = model.predict(student)
# OUTPUT
if result[0] == 0:
print("Behavior Status: NORMAL")
else:
print("Behavior Status: SUSPICIOUS")
```

```
OUTPUT:
Enter number of students: 4
ENTER DETAILS OF STUDENT 1
Average time per question: 50
Number of tab switches: 1
Mouse movements count: 120
Answer submission time: 30
Behavior Status: NORMAL
ENTER DETAILS OF STUDENT 2
Average time per question: 15
Number of tab switches: 10
Mouse movements count: 10
Answer submission time: 3
Behaviour Status: SUSPICIOUS
```

CHAPTER 10 : CODE USED

```
ENTER DETAILS OF STUDENT 3
Average time per question: 34
Number of tab switches: 15
Mouse movements count: 20
Answer submission time: 6
Behavior Status: SUSPICIOUS
ENTER DETAILS OF STUDENT 4
Average time per question: 5
Number of tab switches: 54
Mouse movements count: 45
Answer submission time: 20
Behavior Status: SUSPICIOUS
```

THANK YOU